ALICE L WALTON SCHOOL OF MEDICINE

Immune Thrombocytopenia

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Adapted from: Immune Thrombocytopenia

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of immune thrombocytopenia.
2. Explain the pathophysiology of immune thrombocytopenia, including its morphology in the blood and bone marrow.
3. Interpret clinical findings to properly diagnose immune thrombocyt[illegible]
4. Outline the management plan of immune thrombocytopenia.

CASE CONNECTION

JM is a 34-year-old female with a history of mild systemic lupus erythematosus (SLE) who comes to clinic because of purple spots on her skin. "Here they are," she says, pointing to her legs and calves. "I've been getting them for the past few months." On further questioning, she reports that she's been having heavier menstrual flow for about 3 months, but she has no other symptoms. There is no family history of bleeding disorders. She currently takes no medications, is not pregnant, and has had no recent infections. Her vital signs are within normal limits. Physical exam is notable for scattered petechiae on her lower extremities. You cannot palpate her spleen. On the complete blood count (CBC) JM's white blood cell (WBC) count and hemoglobin are normal and the total platelet count is low.

How will you explain JM's condition? Consider your answer as you read, and we'll revisit JM at the end of the brick.

GO TO CONCLUSION ↓

What Is Immune Thrombocytopenia?

Thrombocytopenia is a condition characterized by abnormally low levels of platelets, typically below 150,000/mm^3. **Immune thrombocytopenia** (ITP), formerly called idiopathic or immune thrombocytopenic purpura, is an acquired form of thro[illegible]uce antibodies that bind to their own platelets, erroneously marking them for destruction in the spleen. The platelet count drops, and eventually there are not enough platelets for normal clotting. ITP can present as a chronic or acute disease but does not usually present with severe bleeding, unlike some other types of thrombocytopenia.

Immune thrombocytopenia is fairly common, with an incidence of 5 to 10 cases/100,000 people in the United States. Historically, it has been considered a disease most common in young females, but more recent data show an increase with advancing age.

Why do platelets decrease in patients with ITP?

How Does a Patient with Immune Thrombocytopenia Present Clinically (LO #1)?

Immune thrombocytopenia has chronic and acute presentations. Common to both are skin abnormalities such as **petechiae or purpura**, and focal or diffuse areas of bleeding within the skin, ranging from discrete red-purple dots to confluent areas (Figure 1, left). Petechiae are smaller (1-3 mm; often described as pinpoint), while purpura are larger. Patients may also present with milder symptoms, such as easy bruising (Figure 1, right).

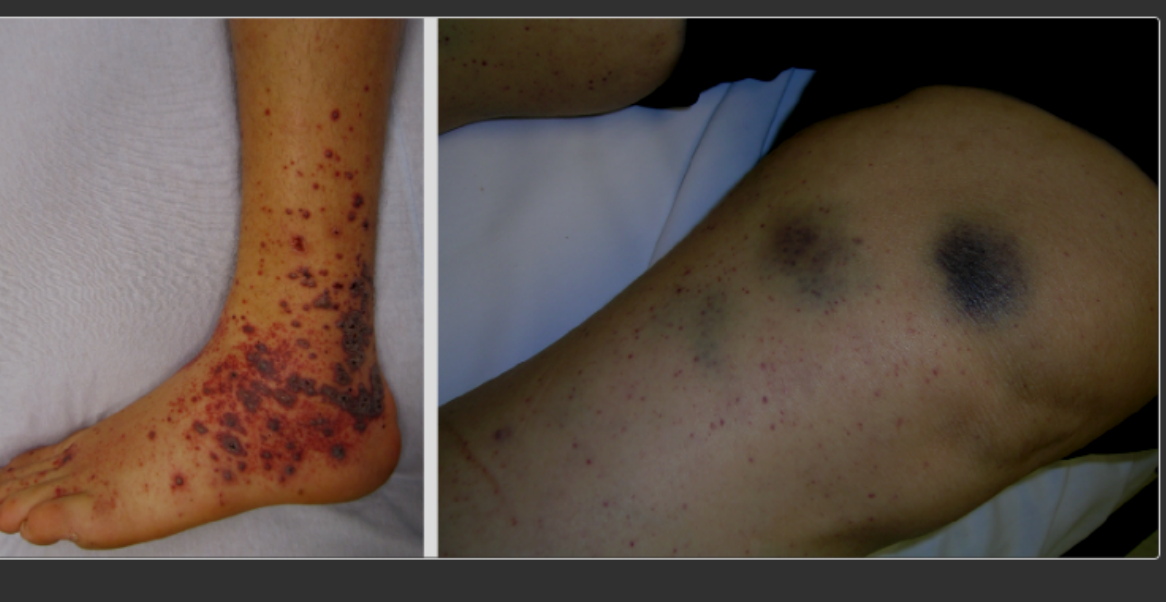

Figure 1

Chronic Immune Thrombocytopenia

Chronic ITP presents most commonly in adult females younger than 40 years. It may be primary (unrelated to any known risk factors) or secondary (occurring in the setting of other conditions, such as lupus, HIV infection, or the hematopoietic neoplasm chronic lymphocytic leukemia). The onset is usually gradual, char[illegible]etechiae, nosebleeds, easy bruising, and excessive menstrual flow. Recurrences are common.

In most cases of chronic ITP, bleeding is relatively minor. Life-threatening bleeding—such as intracerebral bleeding or uncontrollable gastrointestinal bleeding—is rare, particularly with appropriate treatment.

Acute Immune Thrombocytopenia

Acute ITP is most common in children and usually follows a viral illness. Although the onset is abrupt, with the sudden appearance of a petechial rash, bruising, or bleeding, the child usually looks and feels well. In children, the disease is self-limited, generally not requiring treatment. Most children completely recover within a few months.

What are the two types of ITP and the typical patient age for each?

What Is the Pathophysiology of Immune Thrombocytopenia (LO #2)?

In ITP, patients make antibodies (usually IgG) that bind to their own platelets (Figure 2).

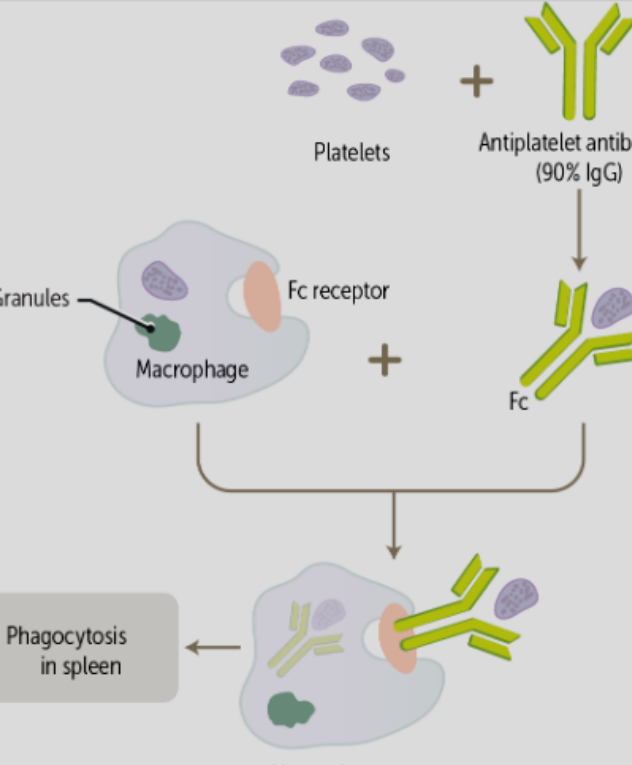

Figure 2

Usually, the antibodies bind to platelet membrane glycoproteins, such as glycoprotein IIb/IIIa. The process of coating cells with antibodies is called opsonization, a word[illegible]ining condiment or delicacy. It's a great word choice, because anything coated with antibodies—even the body's own platelets—looks particularly delicious to macrophages.

In ITP, the macrophages in the spleen gobble up the antibody-coated platelets via interaction with Fc receptors, removing them completely from the circulation. This leads to the reduced platelet number.

Causes

Why would the body make antiplatelet antibodies (antibodies against its own platelets)? Many cases are idiopathic (no identified cause). Some secondary causes include:

- **Viral illness:** the antibodies generated to fight the virus cross-react with the patient's platelets (this is called molecular mimicry). Examples include HIV, hepatitis C, CMV, varicella, and corona virus.
- **Antiviral vaccines (e.g., mumps, measles, rubella):** cases have been reported after COVID-19 vaccination, but it's not clear if there is an actual increased incidence with vaccination.
- **Autoimmune disease:** included are conditions such as lupus and antiphospholipid syndrome, in which normal immune processes have gone awry, and antibodies are made against all sorts of "self-antigens," including those on platelets.
- **Chronic lymphocytic leukemia/small lymphocytic lymphoma:** one of the peripheral B-cell neoplasms.

How do the antiplatelet antibodies damage the platelets?

Histopathology

The peripheral blood smear in a patient with ITP shows an **isolated thrombocytopenia**. This means that the platelets are decreased in number, but everything else—red blood cells and white blood cells—looks normal. The thrombocytopenia may range from very minor (e.g., 100,000-149,000/mm^3) to very severe (e.g., <10,000/mm^3). Occasionally, some of the platelets may be larger than normal, but otherwise, the platelets do not appear abnormal.

The bone marrow in ITP shows **increased megakaryocytes**, as expected given the peripheral loss of platelets. A bone marrow biopsy is not required for diagnosis, however.

What does the blood smear look like in ITP?

How Do We Diagnose Immune Thrombocytopenia (LO #3)?

Immune thrombocytopenia should be considered in any patient with isolated thrombocytopenia but is then diagnosed based on clinical suspicion. For example, are any of the listed secondary causes present, e.g., autoimmune disease or recent viral infection? There are **no specific laboratory tests or diagnostic criteria** that can confirm an ITP diagnosis.

If there are no specific diagnostic criteria, why not just look for the antiplatelet antibodies to prove the diagnosis? It turns out that the antibody tests have a very low sensitivity (about half of the patients with ITP have a negative test result). This makes the test not very useful for diagnosis.

As a result, ITP is considered in patients with an isolated thrombocytopenia (the red blood cells and white blood cells must be normal in number). And to make the diagnosis, you must rule out other causes of the thrombocytopenia. Said another way, ITP is a **diagnosis of exclusion**.

What are the conditions to exclude? The list includes:

- Pseudothrombocytopenia (a falsely low platelet count due to problems in the blood draw, e.g., an insufficiently anticoagulated specimen)
- Hypersplenism (a big spleen can "trap" platelets, making the peripheral count appear low)
- Pregnancy-related thrombocytopenia
- Medication-related thrombocytopenia

Consider ITP when the patient also has one of the secondary causes listed above, such as viral infection or autoimmune disease.

Why are antiplatelet antibodies not used to confirm the diagnosis of ITP?

How Do We Treat Immune Thrombocytopenia (LO #4)?

The goal of treatment is to keep the platelet count at a safe level to prevent severe bleeding. Treatment is typically given to patients with severe bleeding or with very low plate[illegible]eatment can include:

- Corticosteroids (e.g., prednisone): Usual first-line therapy
- Rituximab (anti-CD20 antibody): Typical first choice for relapse after steroid taper
- TPO agonists (e.g., eltrombopag, romiplostim): can be used for recurrence after steroid taper
- Splenectomy: previously standard therapy for corticosteroid failure but now replaced by rituximab or TPO agonists; effective but used 3rd line. Remember to immunize against encapsulated bacteria!
- Intravenous immune globulin (IVIG): used to **rapidly** raise the platelet count but effect is transient

Generally, platelet transfusions are **avoided** in ITP; the transfused platelets are destroyed just as fast as the patient's own platelets. However, they may be used in patients with severe thrombocytopenia and significant bleeding.

CLINICAL CORRELATION

There is not a great correlation between the platelet count and risk of bleeding in ITP; in general, patients with ITP may tolerate a lower platelet count without significant bleeding. In chronic ITP, the goal is to prevent bleeding, not to maintain a normal platelet count.

If treatment fails, splenectomy may be required if patients are symptomatic, in order to remove the source of platelet destruction.

Why would a patient with refractory ITP require splenectomy?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JM, how do you explain her condition?

At this point, given the isolated thrombocytopenia, history of lupus (an autoimmune disease), and lack of other secondary causes of thrombocytopenia (e.g., medications, viral infections, leukemia, drugs), you are convinced that JM has immune thrombocytopenia, likely secondary to her lupus. You explain to JM that ITP is an immune disorder in which the blood doesn't clot normally because the platelet count is too low. "Why does that happen? Am I not making enough of those platelets?" JM asks. You explain that she makes enough platelets, but because of her lupus her body produces antibodies to them and they get destroyed. You refer her to a rheumatologist to consider treatment of her lupus, which should resolve the thrombocytopenia.

Summary

- Immune thrombocytopenia (ITP) is a condition where patients produce antibodies that bind to their own platelets, leading to their destruction in the spleen and resulting in abnormally low levels of platelets.
- ITP can present as a chronic disease, commonly in adult females younger than 40 years, or as an acute disease, mostly in children, usually following a viral illness.
- Immune thrombocytopenia presents with petechiae or purpura or milder symptoms, such as easy bruising.
- In ITP, antibodies (usually IgG) bind to platelet membrane glycoproteins, marking them for destruction by macrophages in the spleen.
- Diagnosis of ITP is based on clinical suspicion and by excluding other causes of isolated thrombocytopenia, as antibody tests have a low sensitivity.
- Treatment of ITP aims to keep the platelet count at a safe level to prevent severe bleeding, and may include corticosteroids, rituximab, TPO agonists, splenectomy, or intravenous immune globulin (IVIG).
- In refractory cases, splenectomy may be required to remove the source of platelet destruction.

Review Questions

1. A 36-year-old female presents with recent onset of unusually heavy menses and occasional nosebleeds. Her physical examination is normal. The complete blood co[illegible]cytopenia. Which of the following, if present in this patient, would best exclude a diagnosis of immune thrombocytopenia (ITP)?

A. ● Blood WBC count of 2000 cells/mm^3 (normal 4500-11,000)
B. ○ Childhood history of meningitis
C. ○ Fasting serum glucose of 150 mg/dL (normal 70-110 mg/dL)
D. ○ Recent viral illness
E. ○ Severe gluten intolerance

Hide Explanation

The correct answer is blood WBC count of 2000 cells/mm^3 (A). To diagnose immune thrombocytopenia (ITP), the thrombocytopenia must first be isolated (meaning there are not any other abnormalities in the complete blood count (CBC), such as a low white blood cell [WBC]). A childhood history of meningitis (B) is irrelevant, as the viral infection is distant, not recent, so would not cause ITP. Diabetes is not a secondary cause of thrombocytopenia, so the elevated serum glucose is irrelevant (C). A history of a recent viral illness (D) would not rule out ITP. In fact, most cases of ITP that occur in children are preceded by a viral illness. A severe gluten intolerance (E) is not relevant to thrombocytopenia.

2. After antiplatelet antibodies attach to platelets, which of the following best explains how the p[illegible]thrombocytopenia (ITP)?

A. ○ The bound antibodies interfere with platelet function
B. ● The platelets are marked for destruction in the spleen
C. ○ The platelets are prevented from leaving the bone marrow
D. ○ The platelets attract NK cells, which destroy the platelets
E. ○ There is activation of complement, leading to intravascular platelet lysis

Hide Explanation

The correct answer is the platelets are marked for destruction in the spleen (B). Antiplatelet antibodies may interfere with platelet function (A), but this question is asking about why the antibodies cause thrombocytopenia, not diminished function. Antibodies are not known to interfere with platelets exiting the marrow (C). Antiplatelet antibodies don't appear to attract NK cells (D) or activate complement (E).

3. In addition to reduced number of platelets, which of the following best describes the peripheral blood smear in a patient with immune thrombocytopenia (ITP)?

A. ○ Irregular, gray platelets
B. ○ Marked leukocytosis
C. ○ Marked red blood cell anisocytosis
D. ○ Megakaryocytes and occasional blasts
E. ● Occasional large platelets

Hide Explanation

The correct answer is occasional large platelets (E). The main finding in immune thrombocytopenia (ITP) is reduced number of platelets. The platelet morphology is either normal or there can be an occasional large platelet. The other cell lines are normal. Irregular, gray platelets (A) suggest gray platelet syndrome, a hereditary condition. Marked leukocytosis (B) suggests the possibility of infection or less commonly leukemia, but not ITP. Marked red blood cell anisocytosis (C) is seen in several anemias (eg, iron deficiency or sickle cell) but not ITP. Megakaryocytes and blasts (D) are made in the bone marrow, so would be seen in the peripheral blood smear most often in patients with bone marrow disorders such as leukemia.

4. Which of the following treatments for ITP rapidly raises the platelet count?

A. ○ Corticosteroids
B. ● Intravenous immune globulin (IVIG)
C. ○ Rituximab
D. ○ Splenectomy
E. ○ Thrombopoietin agonists

Hide Explanation

The correct answer is IVIG (B). This medication can be used to rapidly raise the platelet count, but the effect is transient. This can be used along with steroids in a patient with significant bleeding or as pre-operative treatment. Corticosteroids (A), rituximab (C), splenectomy (D), and thrombopoietin agonists (E) are all also used as treatments for ITP but do not have a rapid effect on the platelet count.

How would you rate this Brick?

References

- Hoffman R, Benz EJ, Silberstein L, et al. *Hematology: Basic Principles and Practice*. 8th ed. Elsevier; 2023.

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of immune thrombocytopenia.~~
02 ☑ ~~Explain the pathophysiology of immune thrombocytopenia, including its morphology in the blood and bone marrow.~~
03 ☑ ~~Interpret clinical findings to properly diagnose immune thrombocytopenia.~~
04 ☑ ~~Outline the management plan of immune thrombocytopenia.~~

Objective Confidence: 4/4